Tabular foundation models for medical prognostic tasks

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Abstract. Survival analysis is a critical task in medical image computing and healthcare informatics, traditionally dominated by statistical methods like CoxPH and specific machine learning models like DeepSurv and Random Survival Forests (RSF). While Tabular Foundation Models (FMs) have shown promise in general classification and regression tasks, their applicability to right-censored survival data remains underexplored. In this work, we investigate whether the representational power of modern Tabular FMs (TabPFN, TabICL, and TARTE) can be leveraged to improve survival predictions. We propose and evaluate two adaptation strategies: (1) extracting learned embeddings from FMs to train downstream survival models, and (2) a Survival Stacking approach using a quantile-based binning strategy to transform time-to-event tasks into classification problems solvable by FMs. We benchmark these methods against five established baselines (CoxPH, DeepSurv, XGBoost, RSF, and Neural Fine-Gray) across three standard datasets (METABRIC, PBC, SUPPORT) and extend our analysis to competing risk scenarios using the SEER and synthetic datasets. Despite rigorous hyperparameter tuning (random and grid search) and 5-fold cross-validation, our experiments reveal that FM-based adaptations do not yield statistically significant improvements over established baselines ($p > 0.05$). These findings suggest that while Tabular FMs capture general data distributions, specialized survival-aware pre-training or architecture modifications are necessary to outperform dedicated survival models.

Keywords: Survival Analysis · Tabular Foundation Models · Benchmarking · Competing Risks.

1 Introduction

Survival analysis, or time-to-event analysis, is ubiquitous in medical research, enabling prognosis estimation and treatment planning. Unlike standard regression, survival analysis must handle censored data, where the event of interest (e.g., death or disease recurrence) has not occurred for all subjects during the study period. While the semi-parametric Cox Proportional Hazards (CoxPH) model remains the clinical standard, it relies on linear assumptions that often fail to capture complex, non-linear interactions in high-dimensional medical data. Consequently, machine learning approaches such as Random Survival Forests (RSF) [3] and deep learning methods like DeepSurv [5] have emerged as powerful alternatives.

Recently, "Foundation Models" (FMs) - large models pre-trained on vast datasets - have revolutionized Natural Language Processing and Computer Vision by enabling few-shot learning and robust transferability. This trend has begun to extend to tabular data, with models such as TabPFN [2], TabICL [8], and TARTE [6] demonstrating the ability to perform classification and regression on unseen datasets with minimal training. However, the potential of these Tabular FMs to enhance survival analysis - a domain often plagued by small sample sizes and high noise - remains largely unverified.

In this study, we hypothesize that the robust feature representations learned by Tabular FMs can improve predictive performance in survival tasks. Since most Tabular FMs are not natively designed for censored regression, we explore two distinct adaptation strategies:

1. **Embedding Extraction:** We utilize TabPFN, TabICL, and TARTE as feature extractors to generate dense embeddings, which are then fed into standard downstream survival models (XGBoost, RSF, DeepSurv, and CoxPH).
2. **Survival Stacking:** We discretize the time horizon into bins and treat survival prediction as a multi-class classification problem. To address class imbalance, we implement a quantile-based binning strategy that ensures an equal distribution of events across time intervals. This allows us to directly apply Tabular FMs as classifiers.

Furthermore, we extend our evaluation to the complex setting of *Competing Risks*, where multiple mutually exclusive events can occur. We investigate Multi-class XGBoost, Neural Fine-Gray (NFG) [4], and a hybrid NFG-embedding approach.

We conducted a comprehensive benchmark using five established baselines (CoxPH, DeepSurv, XGBoost, RSF, and Single-Risk NFG) on standard medical datasets (METABRIC, PBC, SUPPORT). For competing risks, we utilized the SEER database and a synthetic dataset from the DeepHit [7] methodology. All experiments were validated using 5-fold cross-validation and statistical significance testing (t-test). Contrary to the prevailing trend of FMs outperforming specialized models, our results indicate that current FM adaptation strategies do not provide a statistically significant improvement over properly tuned baselines. We provide a detailed analysis of these negative results, discussing the limitations of casting survival problems as standard classification tasks and identifying the gaps preventing current Tabular FMs from generalizing to censored data.

While individual baseline models offer distinct advantages - such as the interpretability of CoxPH or the non-linear capacity of DeepSurv - no single model universally dominates across all data distributions. To leverage the complementary strengths of these diverse models, we implemented a meta-learning approach known as *Stacked Generalization* [9].

Applying stacking to survival analysis introduces a key challenge: base models produce heterogeneous outputs (e.g., hazard ratios, survival probabilities, or

continuous risk scores), which cannot be directly aggregated by a standard classifier. To resolve this, we adopted a *discrete-time survival analysis* formulation [1]. By partitioning continuous time into discrete intervals, the time-to-event task is mapped to a binary classification problem. A more mathematical formulation is provided in Appendix B.

Standard survival analysis assumes a single event of interest (e.g., death). In many clinical settings, however, multiple mutually exclusive outcomes may occur (e.g., cardiovascular death vs. cancer death). Observing one event prevents the occurrence of the others; this is known as *competing risks*. Treating alternative events as simple censoring leads to biased risk estimates. Instead, the clinically relevant quantity is the cause-specific *Cumulative Incidence Function (CIF)*, defined as

$$CIF_k(t) = P(T \leq t, \text{event} = k),$$

which accounts for the possibility that other event types may occur first. The classical statistical solution is the Fine-Gray model [4], which models the subdistribution hazard associated with the CIF. Implementation details and censoring adjustments are provided in Appendix C.

2 Experiments

2.1 Datasets

To rigorously evaluate the proposed tabular foundation model adaptation strategies, we utilized five datasets with diverse sample sizes, dimensionality, and censoring rates. The datasets are divided into *single-risk* and *competing-risk* scenarios.

For standard right-censored survival analysis, we selected three established clinical benchmarks. The **METABRIC** (Molecular Taxonomy of Breast Cancer International Consortium) dataset contains clinical covariates for predicting breast cancer survival time. The **PBC** (Primary Biliary Cirrhosis) dataset, collected by the Mayo Clinic between 1974 and 1984, tracks patients with a rare liver disease to predict mortality. Finally, the **SUPPORT** (Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments) dataset is a large multicenter study of severely ill hospitalized adults, characterized by a higher event rate and missing data which we imputed using mean–mode imputation.

To evaluate the models' ability to distinguish mutually exclusive events, we used two competing-risk datasets. From the **SEER** (Surveillance, Epidemiology, and End Results) database, we extracted a breast cancer cohort where endpoints were defined as death from breast cancer (Event 1) versus death from cardiovascular causes (Event 2). Additionally, we generated a **Synthetic** dataset following the DeepHit methodology, defining non-linear feature interactions mapped to two competing events with a known ground-truth cumulative incidence function.

Table 1. Summary of dataset characteristics

Dataset	Clinical Context	Task Type	N	d	Censored	Event Distribution
METABRIC	Breast Cancer	Single	1,904	9	42.1%	Event: 57.9%
PBC	Liver Cirrhosis	Single	276	17	60.1%	Event: 39.9%
SUPPORT	Severe Illness	Single	9,105	14	31.9%	Event: 68.1%
SEER	Cancer Registry	Competing	39,949	15	70.0%	E1: 15.0%, E2: 15.0%
Synthetic	DeepHit Baseline	Competing	30,000	–	–	2 Events

2.2 Implementation Details

All models were evaluated within a unified benchmarking pipeline to avoid library-specific biases. Survival analysis libraries differ in preprocessing, target encoding, and evaluation procedures, which can otherwise introduce artificial performance differences. Therefore, all methods were trained using identical preprocessing, data splits, and metric implementations. The evaluated models include CoxPH, RSF, XGBoost-Cox, DeepSurv, and Neural Fine-Gray, each executed under the same cross-validation protocol.

Metrics. We evaluate performance using:
- Concordance Index (C-index)
- Integrated Brier Score (IBS)

Training Protocol.
- Stratified 5-fold cross-validation
- Randomized hyperparameter search
- Deterministic seeding across libraries

The goal is to ensure that performance differences reflect modeling capability rather than implementation artifacts.

For the meta-learner in our survival stacking framework, we utilized XGBoost. To prevent data leakage during the meta-learner training phase, we generated the base model predictions $\hat{M}(x_i)$ using strict K-fold cross-validation. Specifically, base models (CoxPH, RSF, DeepSurv) were trained on $K-1$ folds and used to predict the hazard scores of the held-out fold. These out-of-fold predictions were then merged with the Tabular Foundation Model embeddings to construct the final feature matrix for the XGBoost meta-learner. A detailed description of the embedding pipeline, optimization adjustments, evaluation protocol, and reproducibility settings is provided in Appendix A.

3 Results

We evaluated all models using 5-fold cross-validation on the METABRIC, PBC, and SUPPORT datasets. Performance is reported using the time-dependent Concordance Index (C-index) at the median follow-up time (q_{50}) for discrimination, and the Integrated Brier Score (IBS) for calibration.

3.1 Baseline Comparisons

To establish a reference performance ceiling, we first benchmarked five standard survival models across all datasets. As illustrated in Figures 1 and 2, no single model universally dominated across all datasets, but several consistent patterns emerged:

- **Lower Bound:** CoxPH provides a reliable performance floor; complex models failing to surpass it typically suffer from overfitting.
- **Sensitivity:** DeepSurv and XGBoost exhibit strong predictive capabilities but demonstrate high sensitivity to hyperparameter tuning.
- **Stability:** Random Survival Forests (RSF) yield the most stable performance across diverse cohorts, likely due to the variance reduction inherent in tree-based ensembles.
- **Dimensionality:** Non-linear models generally outperform CoxPH on high-dimensional datasets but remain comparable on simpler, lower-dimensional datasets.

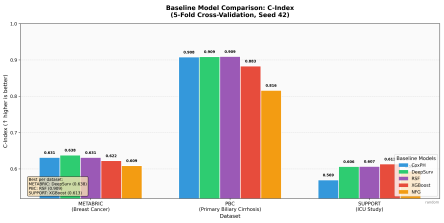


Fig. 1. Baseline comparison using Concordance Index (higher is better).

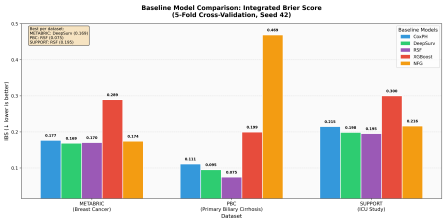


Fig. 2. Baseline comparison using Integrated Brier Score (lower is better).

3.2 Embedding-Based Representation Results

Building upon the baseline analysis, we investigated whether substituting raw features with high-dimensional embeddings extracted from Tabular Foundation Models (TabPFN, TabICL, TARTE) could improve prognostic accuracy.

Figures 3 and 4 detail the discrimination-calibration trade-offs on the METABRIC and PBC datasets. Across all evaluated cohorts, embedding representations did not yield statistically significant improvements ($p > 0.05$) over the raw feature baselines. While the embeddings successfully preserved predictive performance - demonstrating that FMs capture the general data distribution - they failed to reveal additional prognostic signals beyond those already captured by well-tuned tree-based or statistical models.

3.3 Survival Stacking Results

Finally, to leverage the complementary strengths of the diverse base models and FM embeddings, we evaluated our proposed discrete-time stacking approach

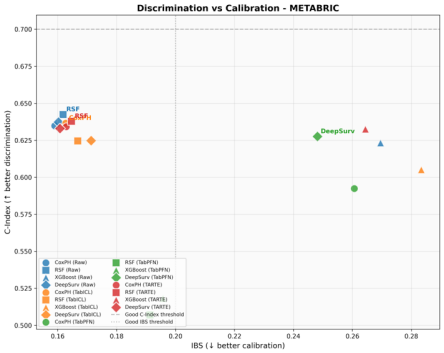


Fig. 3. Discrimination vs. calibration on METABRIC.

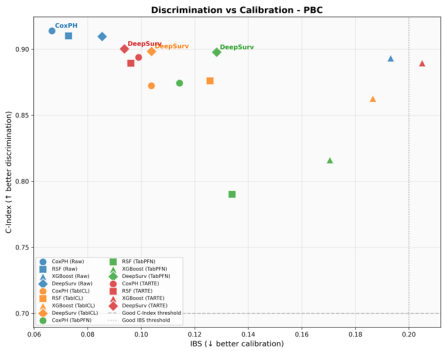


Fig. 4. Discrimination vs. calibration on PBC.

(Figures 5 and 6). Stacking with TabPFN embeddings demonstrated the most consistent and well-calibrated behavior across datasets, though overall improvements over strong baselines remained modest. Dataset-specific behaviors included:

- **METABRIC:** The stacking ensemble achieved the best overall performance, indicating that FM embeddings provided complementary signals to the base models.
- **PBC:** Classical CoxPH and DeepSurv remained superior, suggesting the dataset's prognostic variance is fully captured by standard functional forms without the need for meta-learning.
- **SUPPORT:** Stacking significantly improved overall calibration and mitigated the worst-case failures of CoxPH and XGBoost, though DeepSurv retained slightly higher discrimination.

To rigorously assess these gains, we performed paired statistical tests (paired t-tests) comparing the time-dependent C-index and IBS of the ensemble against the strongest individual base models. While stacking successfully enhanced robustness and stabilized predictions, it did not significantly outperform well-tuned RSF or XGBoost baselines ($p > 0.05$). This indicates that dataset noise and intrinsic clinical signal quality - rather than insufficient model capacity - form the primary bottleneck for performance in these tasks.

4 Discussion and Conclusion

The introduction of our unified experimental framework enabled a rigorous, fair comparison across statistical, machine learning, and deep learning survival models. Our baseline results confirm that evaluation inconsistencies across disjointed libraries often masquerade as algorithmic superiority. We found that tree-based ensembles (e.g., RSF) exhibit the most stable cross-cohort performance, while neural and boosting approaches demand rigorous hyperparameter optimization.

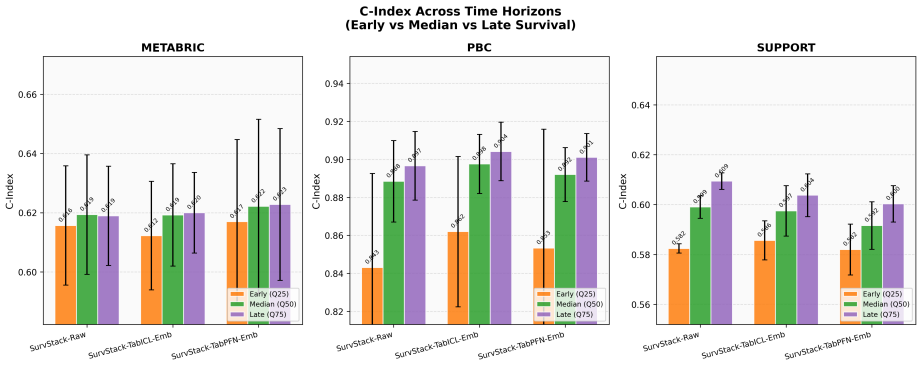


Fig. 5. Time-dependent performance comparison across datasets.

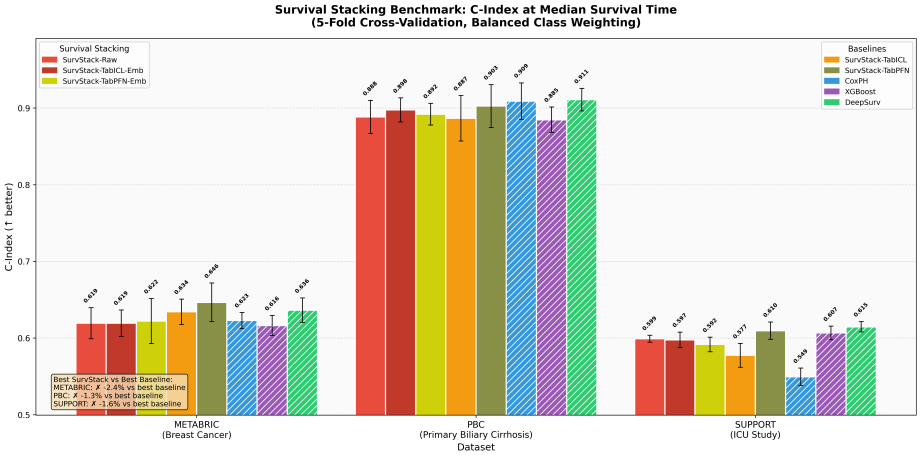


Fig. 6. Overall model comparison using C-index and IBS metrics.

Furthermore, the classical CoxPH model remains a formidable baseline; we argue that the adoption of complex models in clinical settings must be justified by statistically significant gains over CoxPH under identical testing conditions.

Why Foundation Models Failed: A Feature-Space Analysis. The central finding of our study is the lack of statistically significant improvement when utilizing Tabular Foundation Models (TFMs) for survival tasks. To understand this failure, we conducted a feature-space analysis using UMAP projections of the extracted TabPFN and TabICL embeddings versus the raw clinical features.

The visual and cluster analyses reveal a fundamental misalignment in inductive biases: while the TFMs successfully cluster patients based on general covariate similarities, they fail to linearly separate the survival manifold. Specifically, the latent space does not effectively stratify high-risk from low-risk patient trajectories, nor does it isolate censored observations from true events.

This representational collapse occurs because current TFMs are pre-trained exclusively on standard classification and regression tasks, optimizing for cross-entropy or mean squared error. They possess no intrinsic mechanism or loss formulation to account for right-censored data or time-to-event dynamics. When TFMs are utilized as static feature extractors (or within a linear probing/stacking setup), the critical temporal relationships and censoring distributions are lost.

Conclusion and Future Work. Our findings suggest that the predictive information in tabular survival datasets relies heavily on explicit clinical variables rather than the generalized latent interactions captured by current TFMs. For foundation models to succeed in prognostic medical tasks, future research must move beyond naive embedding extraction. We hypothesize that true performance breakthroughs will require survival-aware pre-training objectives—such as embedding Cox partial log-likelihood, time-dependent concordance, or IPCW weighting directly into the self-supervised pre-training loss—enabling these models to natively understand censored risk estimation.

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A Methods

A.1 Feature Representation Learning via Tabular Foundation Models

Theoretical Premise: Manifold Hypothesis Beyond conventional feature engineering, we investigate a representation learning approach using Tabular Foundation Models (TFMs). The underlying assumption follows the manifold hypothesis: high-capacity pre-trained models learn a generalizable mapping

$$x \in \mathbb{R}^d \to z \in \mathbb{R}^D, \quad D \gg d \tag{1}$$

where noisy tabular observations are projected onto a smoother latent manifold. We evaluate three TFMs: TabPFN, TabICL, and TARTE.

We hypothesize that the latent space improves linear separability of survival risk, enabling better modeling of non-linear censoring effects - particularly in small medical datasets where foundation models mitigate limited data.

Embedding Extraction Pipeline We implement a dedicated pipeline treating TFMs as feature extractors rather than final predictors. Unlike classical pipelines that operate on normalized arrays, we provide raw semantic inputs (e.g., age values rather than scaled representations) to preserve distributional context learned during pre-training.

We evaluate three representation modes:
- **Raw:** Original covariates
- **Deep:** High-dimensional embeddings extracted from the TFM
- **Deep+Raw:** Concatenation of embeddings with original features

Each representation is evaluated using the same survival models as the baseline benchmark:
- CoxPH
- RSF
- XGBoost
- DeepSurv

Context-Aware Optimization Training neural models on high-dimensional embeddings required adapted optimization settings. We observed instability when using default hyperparameters due to the variance structure of the learned feature space.

Therefore, we apply context-aware hyperparameter tuning by reducing learning rates $\eta \in \{10^{-4}, 10^{-5}\}$ for embedding-based experiments.

Temporal Quantile Evaluation Instead of relying solely on aggregate metrics, we evaluate discrimination at specific time horizons $q_{25},\ q_{50},\ q_{75}$.

This temporal evaluation determines whether embeddings improve short-term or long-term prognosis prediction.

A.2 Rigorous Evaluation Protocol

Cross-Validation Strategy We employ stratified K-fold cross-validation. In survival analysis, stratification must preserve the ratio of censored to uncensored samples across folds. This ensures validation scores approximate generalization performance.

Hyperparameter Optimization We use **Randomized Search** instead of grid search. Random search is more efficient in high-dimensional hyperparameter spaces because it explores critical dimensions more thoroughly while avoiding redundant evaluations.

Reproducibility To guarantee deterministic experiments, we enforce consistent seeding across NumPy, PyTorch and Scikit-learn. This removes stochastic variation as a confounding factor in benchmarking.

B Details on Survival Stacking

Mathematical Formulation: The Person-Period Transformation. Let the continuous time horizon be partitioned into J discrete intervals defined by cut-points $0 = \tau_0 < \tau_1 < \cdots < \tau_J$. The j-th interval is defined as $[\tau_{j-1}, \tau_j)$.

For a dataset of N independent patients, each patient i is represented by a feature vector x_i, an observed time $T_i = \min(t_i, c_i)$ (where t_i is the true event time and c_i is the censoring time), and an event indicator $\delta_i \in \{0, 1\}$. We transform this cross-sectional data into a longitudinal "person-period" format. Patient i contributes a row for every interval j in which they are at risk (i.e., $T_i \geq \tau_{j-1}$). Let J_i denote the total number of intervals patient i enters.

The binary target indicator $y_{i,j}$ for patient i in interval j is defined as:

$$y_{i,j} = \begin{cases} 1, & \text{if } \tau_{j-1} \leq T_i < \tau_j \text{ and } \delta_i = 1 \\ 0, & \text{otherwise} \end{cases}$$

For each interval, the expanded feature vector $\tilde{x}_{i,j}$ is constructed by concatenating the base model predictions $\hat{M}(x_i)$, the foundation model embeddings z_i, and the temporal context (e.g., the interval index j).

Meta-Learner Objective. With the data expanded into $\mathcal{D}_{exp} = \{(\tilde{x}_{i,j}, y_{i,j})\}$, the stacking problem is reduced to standard binary classification. The meta-learner, parameterized by θ, learns to predict the discrete hazard probability $h_{i,j} = P(y_{i,j} = 1 \mid \tilde{x}_{i,j})$. The model is trained to minimize the regularized binary cross-entropy loss:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{N}\sum_{j=1}^{J_i} [y_{i,j}\log(h_{i,j}) + (1 - y_{i,j})\log(1 - h_{i,j})] + \Omega(\theta)$$

where $\Omega(\theta)$ represents the structural regularization term of the chosen meta-learner to prevent overfitting on the stacked predictions.

C Details on Competing Risks Modeling

Implementation We extended our discrete-time person–period framework to a multiclass formulation, where each interval predicts either survival or a specific event type. We evaluated three approaches:

- **Multiclass XGBoost:** The expanded dataset is treated as a standard classification problem with a softmax objective. Predicted interval probabilities are converted into continuous CIF curves.
- **Neural Fine–Gray:** A neural generalization of the Fine–Gray estimator that models the subdistribution hazard using flexible function approximators.
- **Hybrid Model:** A combined architecture integrating a Cox-like linear baseline with neural residual components, merging their logits before the final prediction layer.

Handling Censoring. To avoid bias toward uncensored observations, we incorporated **Inverse Probability of Censoring Weighting (IPCW)** during training and evaluation. Given the known instability of IPCW when estimating weights from high-dimensional covariates under heavy censoring, we utilized marginal (covariate-independent) censoring probabilities. The censoring survival distribution, $\hat{G}(t) = P(C > t)$, is estimated non-parametrically using a Kaplan–Meier estimator fitted on the training set by treating the censoring indicator as the event of interest.

For an individual i experiencing an event at observed time T_i, the IPCW sample weight is defined as:

$$W_i = \frac{1}{\hat{G}(T_i)}$$

By re-weighting the observations accordingly, the models learn unbiased risk estimates. This marginal approach ensures that evaluation metrics—such as the cause-specific C-index and multiclass IBS—remain reliable without introducing the high variance and instability typical of conditional, Cox-based censoring models.